# Md Jawad Bin Rouf

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LinkedIn | ResearchGate | Google Scholar | Portfolio

Dhaka, Bangladesh

## Education

- **Bachelor of Science in Industrial and Production Engineering** *December 2020 - February 2025*
  *Ahsanullah University of Science and Technology (AUST)* Dhaka, Bangladesh
  - **CGPA: 3.97**/4.0; **Rank:** $1^{st}$ out of **87** students ***(Gold Medalist)***

## Experience

- **Faculty (Lecturer)** *June 2025 - ongoing*
  *Industrial and Production Engineering, AUST*
  - **Conducting Classes:** Preparing lesson plans for undergraduate students, assigning homework, grading tests, and documenting progress for various courses of industrial and production engineering curriculum.
  - **Curriculum & Laboratory Improvement:** Contributed to curriculum development aligned with Outcome-Based Education (OBE) and participated in laboratory improvement initiatives.

- **Research Assistant** *April 2024 - June 2025*
  *Industrial and Production Engineering, AUST*
  - **Project:** Enhancing Occupational Safety Ergonomics in Bangladesh's Ready-Made Garments (RMG) Industry: Multisensory Workload Analysis with Machine Learning (ML) Algorithm
    **Principal Investigator:** Mr. Shah Murtoza Morshed
    * Conduct RMG factory visits, coordinate with industry stakeholders, and collect real-time workload and ergonomic data; Assist in setting up and maintaining multisensory systems; Process data for risk assessment; Manage project documentation and ensure compliance with ethical standards.

## Research Interests

- Applied Machine Learning
- Operations Research and Decision Science
- Predictive and Prescriptive Analytics
- Energy and Sustainable Industrial Systems
- Reinforcement Learning
- Socio-technical Systems
- Human Factors and Occupational Ergonomics
- Human-Centered Artificial Intelligence

## Publications

J=Journal Article, C=Conference Paper, S=Submitted

**[J.1]** **M.J.B.Rouf** et al. **"Evaluating the drivers behind the adoption of end-of-life vehicle management systems in Bangladesh: an intuitionistic fuzzy analytics approach"** *Waste Management Bulletin*, vol.3, Sept.2026.
doi: 10.1016/j.wmb.2026.100318

**[J.2]** M.M.I. Mahin, **M.J.B. Rouf** et al. **"Multivariate time series energy forecasting using a hybrid neural-SVM ensemble model: A data-driven approach for energy management in Bangladesh"** *Energy Nexus*, vol.21, Mar. 2026.
doi: 10.1016/j.nexus.2026.100641

**[J.3]** M.M.I. Mahin, **M.J.B. Rouf** et al. **"An intuitionistic fuzzy analytics approach for examining factors in decentralized renewable energy adoption in emerging economies"** *Decision Analytics Journal*, vol.18, Mar.2026.
doi: 10.1016/j.dajour.2025.100662

**[J.4]** F.B. Mahmud, **M.J.B. Rouf** et al. **"Optimization of rice supply chain through multivariate forecasting using ensemble learning models"** *Supply Chain Insider*, vol.14, Oct.2024.
doi: 10.5281/zenodo.13865817

**[C.1]** M. Morshed, F.B. Mahmud, **M.J.B. Rouf** et al. **"Multivariate Jute Forecasting Using Ensemble Learning for Supply Chain Optimization"** *International Conference on Mechanical, Industrial and Energy Engineering*, vol.3, Jan.2025.
doi.org/10.38032/scse.2025.3.19

**[C.2]** **M.J.B. Rouf** et al. **"Multivariate equipment outage forecasting in the substations of Cumilla grid circle"** *International Conference on Mechanical Engineering and Applied Sciences*, Jul.2025. *Accepted*, Abstract

**[S.1]** **M.J.B. Rouf** et al. **"Assessing Drivers of Psychosocial Stressors using Pythagorean Fuzzy Analytics in Ready-made Garments Industries of Emerging Economies"** *under Review*, Abstract

## Research Experiences

- **Thesis: Performance Evaluation of Machine Learning Algorithms for Multivariate Energy Forecasting: Towards Effective Energy Management in Bangladesh** *September 2024*
  ***Supervisor:*** *Prof. Dr. Mohammad Sarwar Morshed;* ***Tools:*** *Jupyter Notebook, Python, Scikit-learn, Pandas, NumPy*
  - Conducted a study evaluating Machine Learning models for time series forecasting of total energy generation, non-renewable generation, fuel costs, and demand in Bangladesh.
  - Focused on improving forecasting for developing power systems to support policymakers and stakeholders in addressing energy crises through data-driven, sustainable energy planning and management.

- **Project: Enhancing Occupational Safety and Ergonomics in Bangladesh's RMG Industry: Multisensory Workload Analysis with Machine Learning Algorithm** *April 2026*
  ***Principal Investigator:*** *Mr. Shah Murtoza Morshed;* ***Tools:*** *biosignalsplux, OpenSignals, Python, Scikit-learn, NumPy*
  - Analyzed ECG, EMG, EDA, and EEG signals from garment-packaging tasks, performing data preprocessing and feature extraction to assess workers' physiological workload.
  - Contributed to machine learning model development and hyperparameter tuning for workload classification, evaluating generalization across workers using leave-one-subject-out cross-validation.

## Academic Projects

- **Workstation Design Project: Ergonomic Enhancements for Welding** *July 2023*
  *Tools: SOLIDWORKS, Microsoft PowerPoint* [R]
  - Designed and proposed ergonomic enhancements to the AUST welding workstation, including improved furniture, lighting, noise control, ventilation, and tool storage to reduce operator strain and increase efficiency.

- **Hand Tool Design Project: Ergonomic Manual Pry Bar** *July 2023*
  *Tools: SOLIDWORKS, Adobe Illustrator* [R]
  - Redesigned a manual Pry bar tool with an adjustable handle, improved grip, and enhanced visibility to reduce physical strain and increase user efficiency, including capabilities for leg-assisted use.

- **Facility Layout Design: AUST Machine Shop Lab Redesign** *May 2023*
  *Tools: SketchUp, SOLIDWORKS, Microsoft PowerPoint* [R]
  - Analyzed and redesigned the AUST Machine Shop laboratory layout to address spatial inefficiencies, workflow bottlenecks, and safety hazards, optimizing equipment placement, space utilization, and operational efficiency.

- **Product Design and Development: Atmospheric Water Generator** *February 2023*
  *Tools: SOLIDWORKS, ANSYS Fluent, Microsoft PowerPoint* [R]
  - Designed and deployed an Atmospheric Water Generator to address coastal water scarcity by extracting atmospheric moisture via dehumidification and applying multi-stage filtration.

## Standardized Test scores

**IELTS:** 7.0 (Reading 7.5, Listening 7.5, Writing 6.0, and Speaking 6.5)

## Technical Skills

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|---|---|
| **Programming Languages** | : Python, MATLAB, C, C++ |
| **Simulation & Modelling** | : Ansys Fluent, Ansys Structural, Simulink |
| **Graphing & Statistics** | : SigmaPlot, OriginPro, Minitab |
| **Design Software** | : SOLIDWORKS, AutoCAD, TinkerCAD, Adobe Illustrator |
| **Data Science & ML** | : Regression, classification, time-series and multivariate forecasting; supervised, ensemble, reinforcement learning; explainable AI; analysis and model development |
| **Libraries & Frameworks** | : Pandas, NumPy, Scikit-learn, Matplotlib, TensorFlow, Seaborn, Keras, PyTorch |
| **Sensors & Acquisition** | : biosignalsplux (ECG, EMG, EDA, EEG), OpenSignals |
| **Others** | : LaTeX, Arduino, Mendeley, Microsoft Office Suite, Adobe Acrobat |

## Honors and Awards

**Khan Bahadur Ahsanullah Gold Medal (Chancellor's Gold Medal)** *May 2025*
*Ahsanullah University of Science and Technology* [🌐]
• Awarded for achieving the highest CGPA in the graduating batch

**Dean's List of Honor** *February 2025*
*Ahsanullah University of Science and Technology* [🌐]
• Awarded for excellent academic results (CGPA above 3.75)

**Full Free Tuition Award** *All Semester*
*Ahsanullah University of Science and Technology* [🌐]
• Awarded to the top 5% student of the batch

**100% Scholarship, ISCEA Prize – Supply Chain Case Competition** *November 2024*
*International Supply Chain Education Alliance* [🌐]
• Awarded for securing a place among the top 6 teams in the competition

**First Runner-up - Supply Chain Research Challenge 4.0** *October 2024*
*Bangladesh Institute of Human Resource Management* [🌐]
• Awarded for presenting the best supply chain research idea and demonstration

**First Place - IPE Product Showcasing** *February 2023*
*Ahsanullah University of Science and Technology* [🌐]
• Awarded for outstanding product prototype design, CAD modeling, and material selection

**Best Paper Award - the "17th Annual Paper Meet"** *January 2023*
*Institution of Engineers, Bangladesh* [🌐]
• Recognized for exceptional paper writing and delivering the best research presentation

## Leadership Experience

• **Faculty Advisor, IEOM Society AUST Student Chapter** *September 2026 - Present*
*Ahsanullah University of Science and Technology* [🌐]
◦ Mentored student executive members in organizing conferences, workshops, and competitions, aligning chapter activities with IEOM objectives and professional standards.

• **Member, Thesis Committee** *November 2025 - Present*
*Ahsanullah University of Science and Technology* [🌐]
◦ Coordinated undergraduate thesis supervisor selection and organized pre-defense and defense sessions, ensuring smooth academic evaluations.

• **Member, AUST IPE Society** *July 2025 - Present*
*Ahsanullah University of Science and Technology* [🌐]
◦ Conducted transparent elections for student leadership positions and assisted student bodies in planning and executing departmental programs and events.

• **Member, Industrial Attachment and Farewell Committee** *June 2025 - November 2025*
*Ahsanullah University of Science and Technology* [🌐]
◦ Coordinated undergraduate industrial attachments with industry partners and organized departmental farewell events for graduating students.

## Volunteer Experience

• **Department Fire Warden** *November 2025 - Present*
*Ahsanullah University of Science and Technology* [🌐]
◦ Oversaw departmental fire safety and evacuation procedures, conducted routine safety checks, and assisted in organizing fire drills for faculty and students.

• **Amateur Radio (Ham) Operator** *August 2024*
*Disaster Relief Operations, Feni, Bangladesh* [🌐]
◦ Provided emergency Ham radio communications during the 2024 Feni floods, relaying information between affected communities and response teams to coordinate rescue and relief efforts when standard networks failed.